

Unit 4

1-Mark Questions

1. What is an Intent in Android?
2. Define explicit intent with an example.
3. What is the purpose of a context menu in Android?
4. What is a Bitmap in Android?
5. Define Geocoder.
6. What does TTS stand for in Android?
7. Name any two internet connectivity libraries used in Android.
8. What is the use of Bluetooth in Android devices?
9. What does NFC stand for?
10. What is the Android NDK used for?

5-Mark Questions

1. Explain the **types of Android menus** with examples.
2. Discuss **explicit and implicit intents** in Android with suitable code examples.
3. What is a **Bitmap**? Explain its **advantages, disadvantages, and optimization techniques**.
4. Describe **Geocoder** and explain how it is used in **navigation, delivery, and ride-sharing** applications.
5. Explain **Android Text-to-Speech (TTS)** and its **real-world applications** in accessibility and smart assistants.
6. Discuss the role of **Bluetooth, Wi-Fi, and NFC** in building connected Android ecosystems with examples.

Unit 3

1-Mark Questions

1. What does JVM stand for?
2. What is the main file type executed by the Dalvik Virtual Machine (DVM)?

3. Define Object-Oriented Programming (OOP)
4. State advantages of Android being open-source.
5. What is mobile malware?
6. Mention one secure coding practice.
7. Define symmetric encryption.
8. What does HTTPS stand for?
9. What is Google Play Protect used for?
10. Name one common mobile network security threat.

5-Mark Questions

1. Explain in detail the **difference between JVM and DVM** based on their functionality, file types, and optimization.
2. Describe the **evolution from JVM to DVM to ART (Android Runtime)** and explain why Android moved from DVM to ART.
3. Explain the architecture of the Android operating system in detail.
4. Describe various types of mobile security threats with real-life examples.
5. Discuss secure coding practices used in mobile application development.
6. Explain symmetric and asymmetric encryption techniques with examples.
7. What is app sandboxing? Describe its role in mobile security.
8. Evaluate the importance of secure network communication in mobile applications.
9. Discuss cryptography techniques used in mobile apps with suitable examples.
10. Explain the importance of biometric authentication and remote wipe features in device security.

Unit 2

1 Mark Questions

1. What is the main goal of software design?
2. Name any two characteristics of a good Software Design Document (SDD).

3. What are the three phases of software design?
4. Define cohesion in software design.
5. What does a structure chart show?
6. What is the main goal of software engineering?
7. Define software maintenance.
8. What does SDLC stand for?
9. Who developed the Spiral Model?
10. Name any one phase of the Requirement Analysis process.

5 Mark Questions

1. Explain the three phases of the software design process in detail.
2. Discuss the importance of modularity and its benefits in software design.
3. Compare and contrast coupling and cohesion with examples.
4. Describe the top-down and bottom-up design approaches along with their advantages and disadvantages.
5. Explain various estimation models used in software design.
6. Explain the different phases of the Software Development Life Cycle (SDLC).
7. Describe the Prototype Model and its advantages and disadvantages.
8. Discuss the major challenges faced in software development.
9. Explain the various requirement elicitation techniques with examples.
10. Compare the Waterfall Model and Incremental Model with their benefits and limitations.

Unit 1

1 Mark Questions

1. What does “WORA” stand for in Java?
2. Name any two primitive data types in Java.
3. What are the four principles of Object-Oriented Programming?
4. Which keyword is used for inheritance in Java?

5. What is the purpose of encapsulation?
6. Define abstraction in Java.
7. What is polymorphism?
8. Mention one type of polymorphism in Java.
9. Give an example of a control statement in Java.
10. What is an interface in Java?

5 Mark Questions

1. Explain the key features of Java that make it platform-independent.
2. Discuss the concept of inheritance in Java with a suitable example.
3. Describe encapsulation in detail and explain how it enhances data security.
4. What is abstraction? Explain its types and rules in Java.
5. Define polymorphism and discuss its types with examples.
6. Differentiate between compile-time and run-time polymorphism with examples.
7. Explain the importance of OOP principles in Android application development.
8. Discuss the different types of inheritance supported in Java.
9. Describe how control statements and loops work in Java with examples.
10. Explain how abstraction and encapsulation contribute to secure and maintainable software design.